

DEXTER FERNANDES

AI Vision Engineer · Bristol, England · +44 7436 007369 · dexterfernandes11@gmail.com · linkedin.com/in/dexterferns · github.com/Dexter-Fernandes

SUMMARY

Computer vision engineer building machine vision and Vision AI systems for industrial sites. Owned production pipelines end to end for BP drilling rigs and John Deere assembly lines, covering dataset annotation, detector training, inference optimisation and deployment to industrial PCs and edge devices running 24/7. Led a four-engineer team through delivery and on-site validation. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end lifecycle of a real-time vision system on BP drilling rigs, from dataset curation and detector training through edge deployment and on-site validation, improving F1 by 15%.
- Built and hardened NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 perception, adding Dockerised services, health checks and safe restarts to cut pipeline crashes by 67%.
- Adapted YOLOv5 detectors in PyTorch for rig-specific classes and benchmarked them against EfficientDet on accuracy, latency and stability, catching regressions before release.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to correct mud and sand contamination on camera feeds, reducing false positives by 13%.
- Built a watchdog service with a REST API that detected and restarted failed edge services, removing the need for site visits to recover pipelines.
- Led on-site validation on rig deployments and presented findings to leadership, project managers and operators, turning field failures into rollout decisions.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector and helmet-colour classifier for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial PCs, raising throughput from 5 to 15 FPS under real-time constraints.
- Built YOLOv5 automated inspection for John Deere engine paintjobs on a conveyor line, flagging paint defects, contamination and colour inconsistency for quality control.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Synchronised a multi-camera pipeline on a moving construction vehicle and tuned camera and ISP settings across RGB and IR streams, cutting false positives and manual recalibration.
- Automated labelling by repurposing existing detection models and added dataset versioning with DVC, cutting manual labelling effort by 80% on large real-world datasets.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream.
- Implemented a C++ and CUDA TensorRT decode plugin, moving post-processing into the inference engine to support custom-layer deployment.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess segmentation and depth models under the same compute constraints.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer fusing wheel odometry and IMU.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Calibrated LiDAR and camera extrinsics to the robot base and resolved transform timing across a PCL point cloud pipeline, validating sensor placement for coverage.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier, working within the platform's compute and memory limits for on-robot operation.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash
Computer vision	Object detection, image classification, semantic and instance segmentation, defect and anomaly detection, object tracking, optical flow, data augmentation, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, U-Net, Mask R-CNN, SegFormer
Deep learning	PyTorch, TensorFlow, Keras, OpenCV, scikit-image, torchvision, NumPy, model training and fine-tuning, hyperparameter optimisation (Optuna), benchmarking and performance evaluation, Weights & Biases
Edge AI	OpenVINO, TensorRT, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, Intel VTune, Intel Movidius VPU, Intel Core i7 industrial PCs, NVIDIA Jetson AGX Xavier
Industrial imaging	Multi-camera synchronisation, ISP and camera tuning (RGB and IR), image acquisition and correction, camera and LiDAR extrinsics calibration, RTSP ingestion, NVIDIA DeepStream, GStreamer, FFmpeg
Data and annotation	CVAT, DVC (Data Version Control), MLflow, Apache Airflow, COCO annotation format, automated labelling, dataset versioning, synthetic data generation, data quality management, drift tracking, TIDE error analysis, Grad-CAM
Software delivery	Docker, Git, CMake, CI/CD, pytest, gtest, REST API design, multiprocessing, Linux, structured logging, production monitoring
Robotics	ROS2 (Robot Operating System 2), tf2, rosbag2, Gazebo, RTAB-Map, Nav2, PCL (Point Cloud Library), Open3D, LiDAR SLAM
Professional	Technical leadership, mentoring, customer-facing delivery, Agile and Scrum, Jira, stakeholder communication

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, deployment and software engineering.

Taco Bell — Service Champion / Team Trainer

09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) · Best Speaker award, ABC Toastmasters Club (01/2020)